

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 18-03-26
Time: 3 Hrs Max. Marks: 300

18-03-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-9_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP

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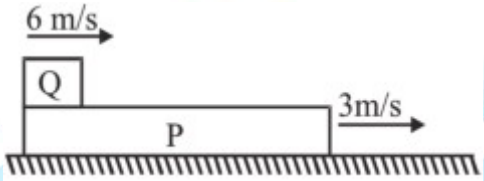
PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

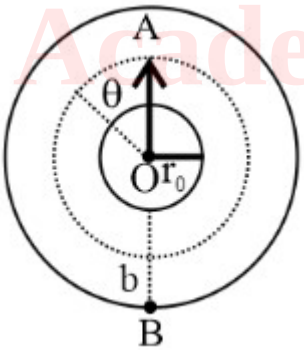
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A plank P and block Q are arranged as shown on a smooth table top. They are given velocities $3\ m/s$ and $6\ m/s$ respectively. The length of plank is 1 m and block is of negligible size. After some time when the block has reached the other end of plank it stops slipping on plank. The velocity of plank then is (coefficient of friction between plank and block is 0.3 and mass of plank is double of block).



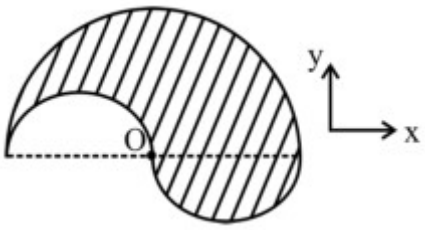
A) $4\ m/s$ B) $5\ m/s$ C) $4.5\ m/s$ D) Zero

2. In a water park there is a circular river in which the flow is at a speed u along a circular path. A small toy beat is steered radially with speed v relative to water. If reaches a diametrically opposite point B to the other bank in half round (180°). The time to cross the river and reach B is. (r_0 is radius of central circle.)



A) $\frac{r_0\left(e^{\frac{v\pi}{u}}-1\right)}{v}$ B) $\frac{r_0\left(e^{v}-e^{\pi}\right)}{v}$ C) $\frac{r_0\left(e^{\pi}-e^{\frac{v\pi}{u}}\right)}{v}$ D) None of these

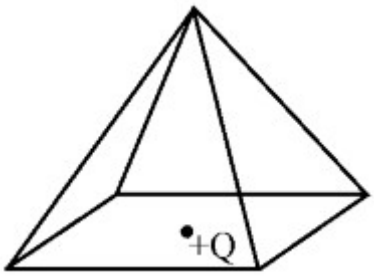
3. From a uniform semicircular thin disc of radius R , a semicircular portion of radius $R/2$ is removed and placed in a new position as shown in figure. Taking O as origin and x and y axis as shown in figure, centre of mass of new setup is at :-



A) $\frac{R}{4},\frac{4R}{3\pi}$ B) $-\frac{R}{4},\frac{R}{3\pi}$ C) $\frac{R}{4},\frac{R}{\pi}$ D) $\frac{R}{4},\frac{R}{3\pi}$

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4. A point charge +Q is positioned at the centre of the base of a square pyramid as shown. The flux through one of the four identical upper faces of the pyramid is :-

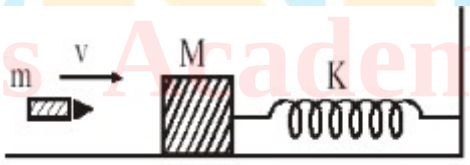


A) $\frac{Q}{16\varepsilon_0}$ B) $\frac{Q}{4\varepsilon_0}$ C) $\frac{Q}{8\varepsilon_0}$ D) None of these

5. A particle of mass m is moving with a uniform velocity v on straight line $3y = 2x + 6$. Two observers A and B are situated at $(0,0)$ and $(x_1,0)$. If the angular momentum of the particle with respect to B is twice the angular momentum with respect to A , find the value of x_1 :

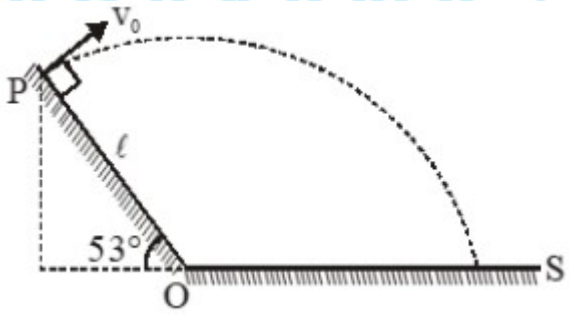
A) 1 B) 2 C) 3 D) 4

6. A block of mass M , at rest on a horizontal frictionless table, is attached to a rigid support by a spring of constant K . A bullet of mass m with velocity v strikes the block as shown in figure. The bullet remains embedded in the block. Then the amplitude of the resulting simple harmonic motion is :-



A) $\frac{mv}{K}$ B) $\frac{Mv}{K(m+M)}$ C) $\frac{mv}{\sqrt{K(m+M)}}$ D) $\frac{Mv}{\sqrt{K(m+M)}}$

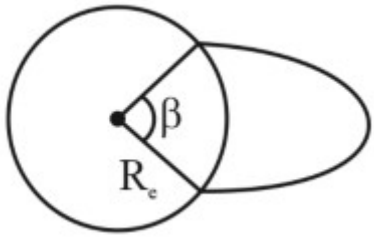
7. A stone is projected from point P on the inclined plane with velocity $v_0 = 10$ m/s directed perpendicular to the plane. The time taken by the stone to strike the horizontal ground S is_____
(Given $PO = \ell = 10$ meter)



A) 1.5 sec B) 1.4 sec C) 2 sec D) 2.3 sec

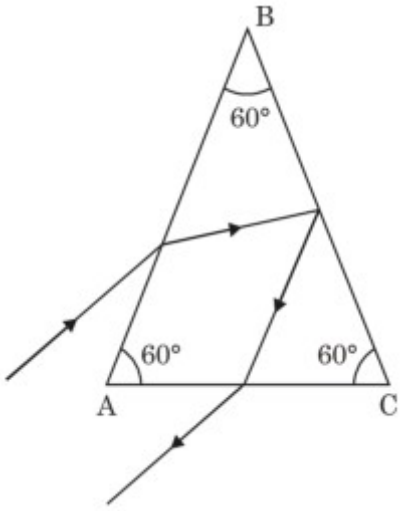
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8. A particle is projected from the surface of the earth (of radius R_e and mass M_e) with speed equal to $\sqrt{\frac{GM_e}{R_e}}$, at certain angle from local horizontal as shown in the figure such that the angle subtended by arc between launching & landing site at earth's centre is $\beta = 60^\circ$. Calculate the maximum separation of the particle from the surface of the earth. Consider earth to be uniformly dense and air resistance to be absent.



A) $R_e\left(1+\frac{\sqrt{3}}{2}\right)$ B) $R_e\left(\sqrt{2}+1\right)$ C) $R_e\frac{\left(\sqrt{2}+1\right)}{\sqrt{2}}$ D) $\frac{R_e\sqrt{3}}{2}$

9. Consider an equilateral prism ABC as shown in the figure. A ray of light is incident on the face AB and gets transmitted through the prism. Then total internal reflection takes place at the face BC and the ray comes out of prism through the face AC. The total angle of deviation is 120°. Minimum value of refractive index m of the material of the prism should be



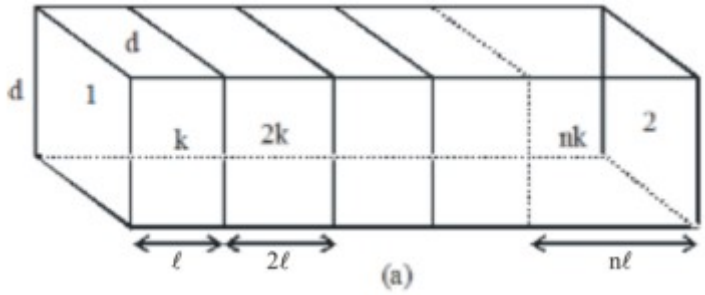
A) $\sqrt{\frac{4}{3}}$ B) $\sqrt{\frac{5}{3}}$ C) $\sqrt{2}$ D) $\sqrt{\frac{7}{3}}$

10. If T is the surface tension of a liquid, the energy needed to break a liquid drop of radius R into 64 drops is :-

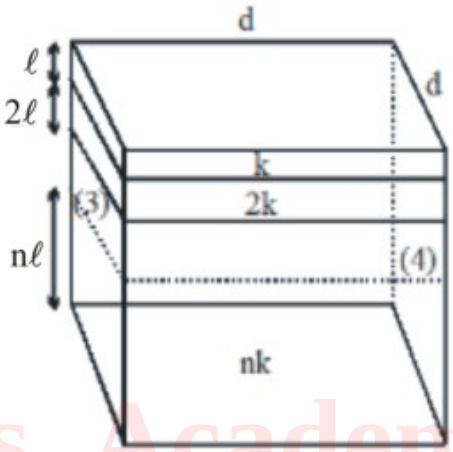
A) $6\pi R^2T$ B) πR^2T C) $12\pi R^2T$ D) $8\pi R^2T$

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11. An arrangement consisting of slabs is shown in the figure. When system is placed in situation-(a) with temperature difference maintained between (1) & (2), equivalent thermal conductivity of system is K and when system is changed to situation-(b) and temperature difference is maintained between (3) & (4) then equivalent thermal conductivity increases by 20%. Find number of slabs in the arrangement.



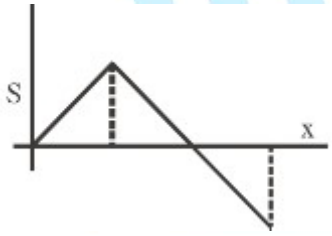
(a)



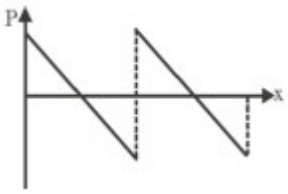
(b)

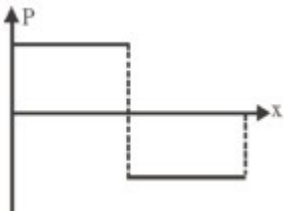
A) 4 B) 5 C) 2 D) 3

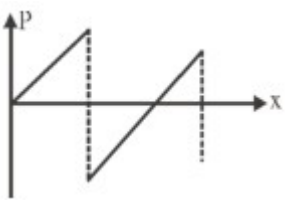
12. A sound wave is propagating through a medium such that displacement $v/s\times$ graph is

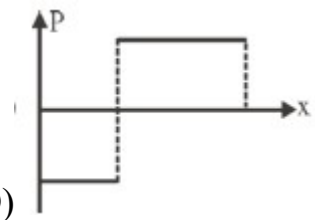


Corresponding excess pressure v/s x graph will be

A) 

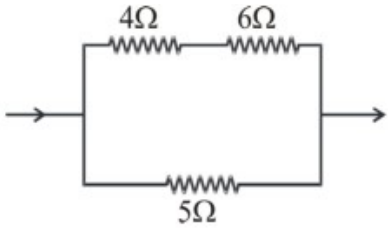
B) 

C) 

D) 

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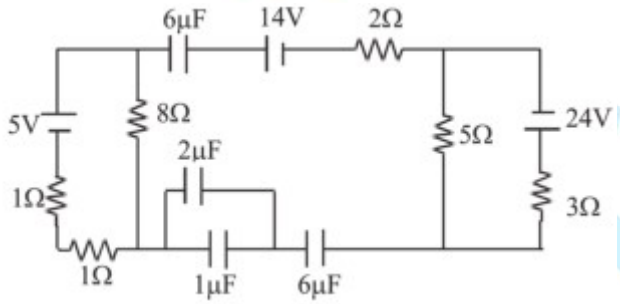
13. In the circuit shown in figure the heat produced in the 5Ω resistor due to the current flowing through it is 10 calories per second.



The heat generated in the 4Ω resistor is :-

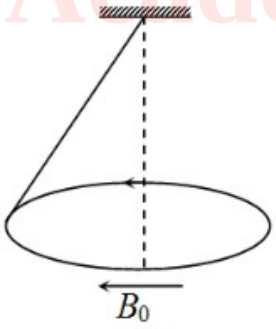
A) 1cal/s B) 2cal/s C) 3cal/s D) 4cal/s

14. Find the charge on the capacitors of value $2\mu\text{F}$ in the shown figure at steady state



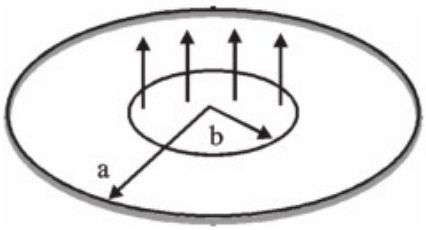
A) $3\mu\text{C}$ B) $5\mu\text{C}$ C) $1.5\mu\text{C}$ D) $2.5\mu\text{C}$

15. A uniform current carrying ring of mass m and radius R is connected by a massless string as shown. A uniform magnetic field B_0 exist in the region to keep the ring in horizontal position, then the current in the ring is (ℓ = length of string)



A) $\frac{mg}{\pi RB_0}$ B) $\frac{mg}{RB_0}$ C) $\frac{mg}{3\pi RB_0}$ D) $\frac{mg\ell}{\pi R^2 B_0}$

16. A line charge (per unit length) is in the form of circular wheel of radius ' a ' and moment of inertia is I, initial at rest. It is free to rotate in a horizontal plane. There is a coaxial magnetic field $\vec{B} = B_0\hat{k}$ extending upto a radius $b(b<a)$. If the magnetic field is switched off, the angular velocity ' ω ' of the wheel is given by



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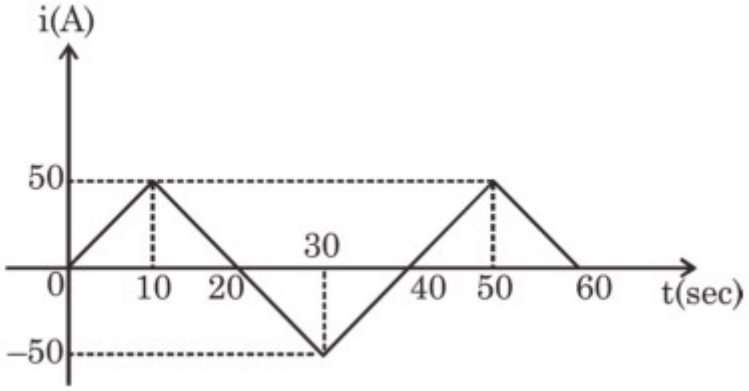
A) $\frac{\pi a^2 b\lambda B}{I}$ clockwise as seen from above

B) $\frac{\pi ab^2\lambda\ B}{I}$ clockwise as seen from above

C) $\frac{\pi ab^2\lambda\ B}{2I}$ anticlockwise as seen from above

D) $\frac{\pi ab^2\lambda\ B}{I}$ anticlockwise as seen from above

17. Value of i_{rms} for the following graph for $t=0$ to $t=60\ s$ is :



A) $50\sqrt{3}$ A B) $\frac{50}{\sqrt{3}}$ A C) 50 A D) None of these

18. A laser gun of power 3 mW and mass 50 gm emits photons of wavelength 500 nm . It is in gravity free space and emits for one hour. Find the distance covered by gun due to recoil in this one hour (approx.) :-

A) 1.3 mm B) 2.1μ m C) 1.1 cm D) 8.5 mm

19. At a point P on the screen at distance D from the slits in standard YDSE experiment, 3rd maxima is observed at $t=0$. Now screen is slowly moved with constant speed away from the slits in such a way that the centre of slits and centre of screen always lie on same line and at $t=1sec$, the intensity at point P is observed to be $\left(\frac{3}{4}\right)^{th}$ of maximum intensity in between 2nd and 3rd maxima. The speed of screen can be :-

A) $\frac{5D}{13}$ B) $\frac{13D}{5}$ C) $\frac{17D}{5}$ D) $\frac{7D}{17}$

20. Two polaroids are placed in the path of unpolarized beam of intensity I_0 such that no light is emitted from the second polaroid. If a third polaroid whose polarization axis makes an angle θ with the polarization axis of first polaroid, is placed between these polaroids, then the intensity of light emerging from the last polaroid will be :-

A) $\left(\frac{I_0}{8}\right)\sin^2 2\theta$ B) $\left(\frac{I_0}{4}\right)\sin^2 2\theta$ C) $\left(\frac{I_0}{2}\right)\cos^2\theta$ D) $I_0\cos^4\theta$

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A vessel, whose bottom is flat and perfectly reflecting, is filled with water (index $=4/3$) upto a height $=40\ \text{cm}$. A point object in air above is moving towards the water surface with a constant speed $=4\ \text{m/s}$. What is the relative speed of its final image (in m/s), as seen by the object itself, at a moment when the object is 30 cm above the water surface?

22. One end of a string of length L is tied to the ceiling of a lift accelerating upwards with an acceleration 2g. The other end of the string is free. The linear mass density of the string varies linearly from 0 to λ from bottom to top. The acceleration of the wave through out string is $\frac{pg}{4}$. Find p.

23. Two blocks of mass 1 kg and 0.5 kg moving in opposite direction with velocity of $0.5\ \text{m/s}$ and $1\ \text{m/s}$ respectively, collide. Coefficient of restitution is $e=\frac{1}{2}$. After collision, the loss in energy is distributed equally between both the blocks. The specific heat capacity of 1 kg block is $1000\ \text{J/kg}^{\circ}\text{C}$ and the change in its temperature (in $10^{-3}\ ^{\circ}\text{C}$) is ΔT. Find $64\Delta T$.

24. In an α-decay the kinetic energy of a particle is 48 MeV and Q value of the reaction is 50 MeV . The mass number of the parent nucleus is 20 n then find n : (Assume that daughter nucleus is in ground state)

25. In a screw gauge, 5 complete rotations of the screw cause it to move a linear distance of 0.25 cm and each rotation corresponds to one main scale division. There are 100 circular scale divisions. The thickness of a wire measured by this screw gauge gives a reading of 4 main scale divisions and 30 circular scale divisions. Assuming negligible zero error, what is the thickness (in $10^{-5}\ \text{m}$) of the wire?

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. What is the value of $pK_b\left(CH_3COO^-\right)$, if $\lambda_m\infty = 390$ and $\lambda_m = 7.8$ for 0.04 M of a CH_3COOH solution at $25^\circ C$?

A) 9.00 B) 9.20 C) 8.57 D) 8.92

27. The total number of possible isomers for squareplanar complex $\left[PtCl(NO_2)(NO_3)(SCN)\right]^{-2}$ is :

A) 10 B) 11 C) 12 D) 13

28. Which of the following does not represent the correct order of the property indicated against it?

A) $F > Cl > Br > I$ (Electronegativity)

B) $Li > B > Be > C$ (Second ionisation enthalpy)

C) $O < S > Se > Te$ (Electron affinity)

D) $Ti < Zr \approx Hf$ (Atomic size)

29. Consider the half cell reactions :

(i) $6OH^- + Br^- \rightarrow BrO_3^- + 3H_2O + 6e^-,\ E^\circ = -0.61\ V$

(ii) $2OH^- + Br^-$ & nbsp $\rightarrow + H_2O + 2e^-,\ E^\circ = -0.76\ V$

Then the value of ΔG° for the following reaction is :

$BrO^- + 4OH^-$ & $nbsp \rightarrow nbspBrO^- + 2H_2O + 4e^-, DG^\circ = ?$

A) -206.5 kJ B) 206.5 kJ C) -106.5 kJ D) 106.5 Kj

30. The transition metal ions responsible for color in ruby and emerald are, respectively :

A) Co^{+3} and Cr^{+3} B) Co^{+3} and Co^{+3}

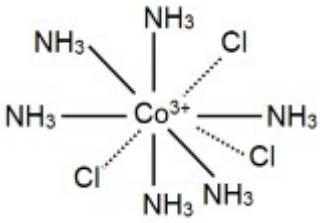
C) Cr^{+3} and Cr^{+3} D) Cr^{+3} and Co^{+3}

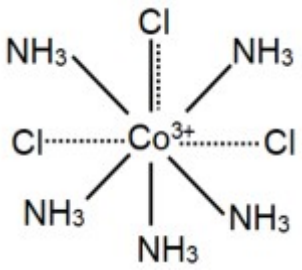
31. In a particular case of bacterial growth following second order kinetics, concentration of bacteria increases to 4 times initial concentration of 1 M in 24 minutes. What will be the generation time of the bacterial growth , if initial concentration is 2 M .

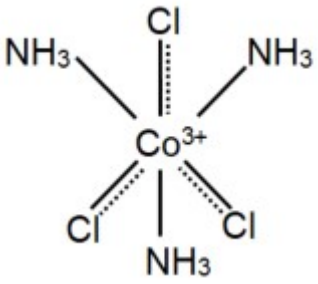
A) 24 min B) 16 min C) 8 min D) 12 min

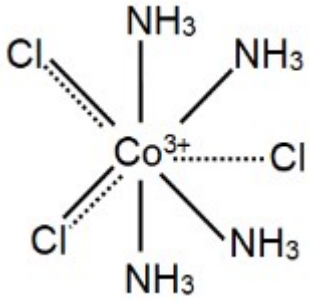
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32. Which of the following Werner's complex has least electrical conductivity?

A) 

B) 

C) 

D) 

33. Identify the pair in which the geometry of the species is T-shape and square pyramidal, respectively?

A) ClF_3 and IO_4^- B) ICl_2^- and ICl_5

C) $XeOF_2$ and $XeOF_4$ D) IO_3^- and $IO_2F_2^-$

34. Which of the following options consist of only those orbitals which have number of radial nodes exactly same as their angular nodes?

A) $1s, 2p, 3d$ B) $1s, 2s, 3s$ C) $1s, 3p, 5d$ D) $2s, 3p, 4d$

35. A magnetic moment of 1.73 BM will be shown by the following :

A) $[Ni(CN)_4]^{2-}$ B) $TiCl_4$ C) $[CoCl_6]^{4-}$ D) $[Cu(NH_3)_4]^{2+}$

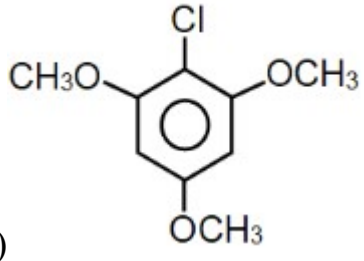
36. For an ideal solution, P_A° and P_B° are the respective vapor pressure of the pure constituents. X_A and X_B are the mole fractions of two constituents in liquid phase. Y_A and Y_B are the mole fractions in the vapour phase. The plot of $\frac{1}{X_A}$ (on y -axis), $\frac{1}{X_A}$ (on x -axis) is linear with slope and intercept respectively :

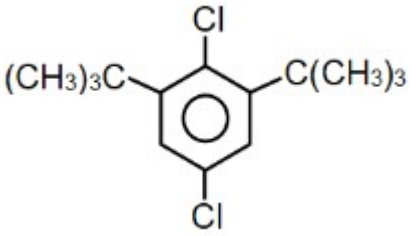
A) $\frac{P_A^o}{P_B^o}$ and $\left(\frac{P_A^o - P_B^o}{P_B^o}\right)$ B) $\frac{P_A^o}{P_B^o}$ and $\left(\frac{P_A^o - P_B^o}{P_B^o}\right)$

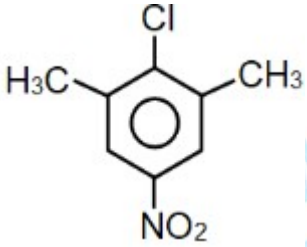
C) $\frac{P_A^o}{P_B^o}$ and $\left(\frac{P_B^o - P_A^o}{P_B^o}\right)$ D) $\frac{P_B^o}{P_A^o}$ and $\left(\frac{P_B^o - P_A^o}{P_B^o}\right)$

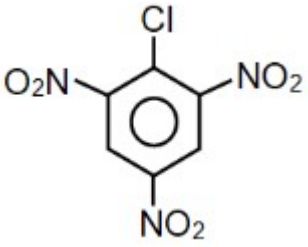
For More Join: @IITJEE_Advanced

37. Which of the following compounds undergoes replacement of -Cl with -OH by nearly warming the compound with aqueous NaOH ?

A) 

B) 

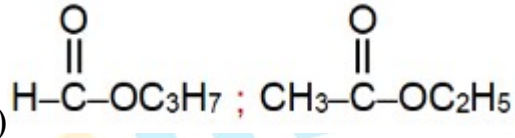
C) 

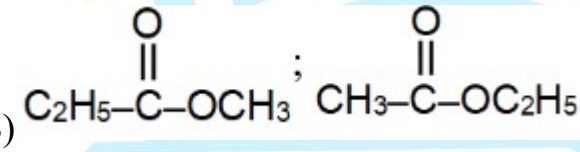
D) 

38. Compound $X(C_4H_8O_2)$ is a sweet smelling liquid it gives negative test with Na metal and $NaHCO_3$.

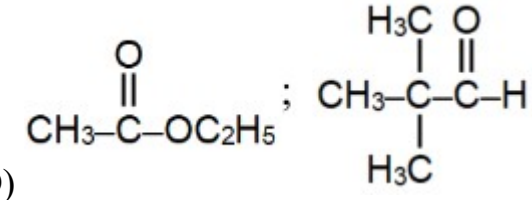
An another isomer $Y(C_4H_8O_2)$ responds positive to 2,4-DNP and iodoform test but negative with $NaHCO_3$.

Identify the structures of X and Y respectively :

A) 

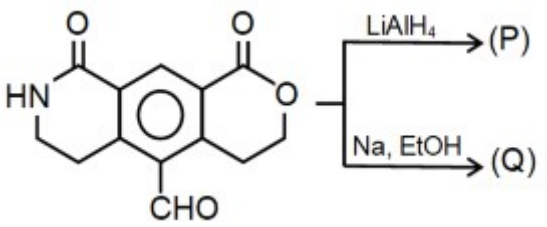
B) 

C) 

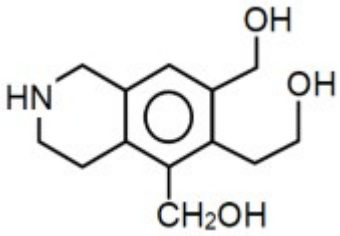
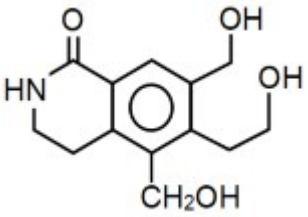
D) 

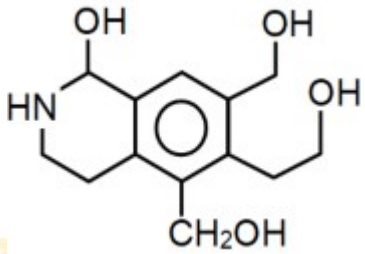
For More Join: @IITJEE_Advanced

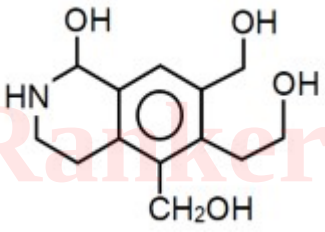
39.

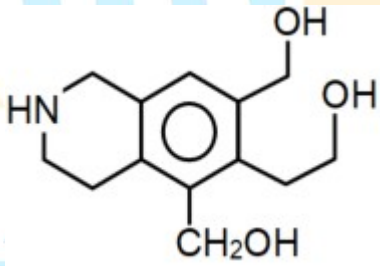


Identify (P) and (Q) respectively

A)  & 

B) Both 

C)  & 

D) Both 

40. Which of the following reaction can be used to prepare isobutane in good yield?

(a) Wurtz reaction of CH_3Br and $CH_3CH_2CH_2Br$.

(b) Hydrolysis of iso-butyl magnesium iodide.

(c) Kolbe electrolysis of potassium salt of ethanoic acid.

(d) Decarboxylation of 3-methylbutanoic acid.

A) a & b B) b & c C) b & d D) c & d

For More Join: @IITJEE_Advanced

41. Which is the incorrect match?

	Compound	Natural source
A)	Vanillin	Vanilla beans
B)	Cinnamaldehyde	Vinegar
C)	Butyric acid	Rancid butter
D)	Salicylaldehyde	Meadow sweet

42. Order of basic strength of the following compounds is -

(a) $p-CH_3-C_6H_4-NH_2$

(b) $Et-NH_2$

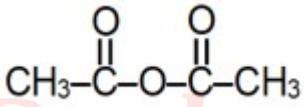
(c) $Ph-NH-CH_3$

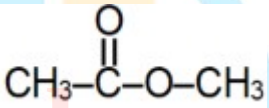
(d) $Ph-N(CH_3)_2$

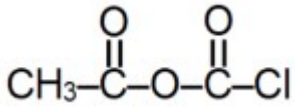
A) b > d > c > a B) b > d > a > c

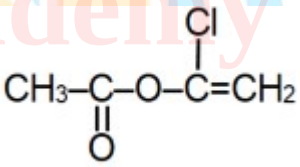
C) a > d > c > b D) a > c > d > b

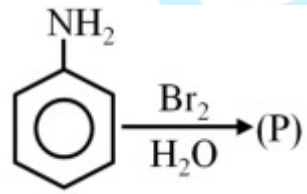
43. $CH_2=C=O \xrightarrow[(2)\ CH_3COCl]{(1)\ H_3O^+}$ product is

A) 

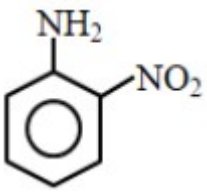
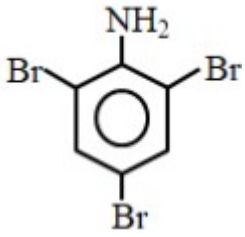
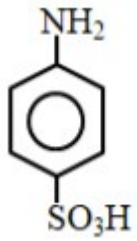
B) 

C) 

D) 

44. 

P is-

A)  B)  C)  D) 

For More Join: @IITJEE_Advanced

45. The correct order of boiling point is :

A) $H_2O < H_2S < H_2Se < H_2Te$

B) $H_2O > H_2Se > H_2Te > H_2S$

C) $H_2O > H_2S > H_2Se > H_2Te$

D) $H_2O > H_2Te > H_2Se > H_2S$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The mole fraction of glucose in an aqueous solution is $\frac{9}{109}$. The molality of solution is:

47. $1dm^3$ of an ideal gas at a pressure of 10 bar expands reversibly and isothermally to final volume of 10 litre. What will be the heat absorbed in the process ? ($\ln 10 = 2.3$)

48. Total number of structural isomers of C_4H_7Cl, which can show geometrical isomerism is?

49. $C_6H_5NO_2 \xrightarrow[Pd]{H_2} Q \xrightarrow[0^\circ C]{HNO_2} R \xrightarrow[boil]{H_2O} S \xrightarrow[H_2O]{Br_2 \text{ excess}} T$ Mass percentage of bromine in the final product T is :

50. A d-block element has total spin value of +3 or -3 , then the number of unpaired electorn is:

For More Join: @IITJEE_Advanced

MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. In triangle $\triangle ABC$, if $\overrightarrow{AB}=\frac{\vec{u}}{|\vec{u}|}-\frac{\vec{v}}{|\vec{v}|}$ and $\overrightarrow{AC}=\frac{2\vec{u}}{|\vec{u}|}$ where $|\vec{u}|\neq|\vec{v}|$, then $\sum\cos(2A)$ equals

A) -2 B) -3 C) -1 D) -4

52. The perpendicular distance of point $(2,0,-3)$ from line which passes through the point $(0,2,-4)$ and perpendicular to the lines $\bar{r}=\left(-3\hat{i}+2\hat{k}\right)+\lambda\left(2\hat{i}+3\hat{j}+5\hat{k}\right)$ and $\bar{r}=\left(\hat{i}-2\hat{j}\right)+\mu\left(-\hat{i}+3\hat{j}+2\hat{k}\right)$, where $\lambda,\mu\in R$ is :-

A) $\frac{\sqrt{219}}{3}$ B) $\frac{\sqrt{78}}{3}$ C) $\frac{\sqrt{52}}{3}$ D) $\frac{\sqrt{126}}{3}$

53. Let $A_1\ A_2....A_n$ be a regular n -sided polygon inscribed in a circle. Three points A_i, A_j, A_k are randomly chosen, where i,j,k are distinct integers between 1 and n , both inclusive. Denote by P_n the probability that the triangle $A_iA_jA_k$ is obtuse. Which of the following statements is/are incorrect?

A) P_5 equals $\frac{1}{2}$ B) P_{10} equals $\frac{1}{2}$

C) P_{10} equals $\frac{2}{3}$ D) P_{15} equals $\frac{9}{13}$

54. Let $P\left(\frac{2\sqrt{3}}{\sqrt{7}},\frac{6}{\sqrt{7}}\right),Q,R$ and S be four points on ellipse $9x^2+4y^2=36$. Let $PQ\ \&\ RS$ be mutually perpendicular chords and these pass through the centre of ellipse then the value of $\left[\frac{50}{PQ^2}+\frac{50}{RS^2}\right]=.....$ where [.] denotes GIF

A) 4 B) 5 C) 6 D) 7

For More Join: @IITJEE_Advanced

55. $\int \frac{x\left(x\tan^{-1}x+(\ln x)(\ln(\ln x))\right)+\tan^{-1}x}{\left(x^3+x\right)\ln x}dx = f(x)+c$ where ' c ' is integration constant and

$f(e)=0$. Then $\left[\lim_{x\to 1^+}\frac{f(x)}{\tan\left(\frac{\pi x}{2}\right)}+\frac{11}{10}\right]=$ ____ where [.] denotes GIF

A) 2 B) 1 C) 0 D) -1

56. Let $y=f(x)$ be a differentiable function satisfying $\int_2^x f(t)dt+2=\frac{x^2}{2}+\int_x^2 t^2 f(t)dt$ then

$\int_{-\pi/4}^{\pi/4}\frac{f(x)+x^9-x^3+x+1}{\cos^2 x}dx=$

A) 2 B) 3 C) 4 D) 0

57. Let ' q ' be the maximum integral value of p in $[0,10]$ for which the roots of the equation $x^2+px+\frac{5p}{4}=0$ are rational then the area of region $\{(x,y):0\leqslant y\leqslant(x-q)^2,0\leqslant x\leqslant q\}$ is (in square units)

A) 243 B) 25 C) $\frac{125}{3}$ D) 164

58. A classroom has a row of 35 coat hooks. Paulina likes coats to be equally spaced, so that there is the same number of empty hooks before the first coat, after the last coat, and between every coat and the next one. Suppose there is at least 1 coat and at least 1 empty hook. How many different numbers of coats can satisfy Paulina's pattern?

A) 2 B) 4 C) 5 D) 7

59. The value of S , where $S=\sum_{(r=0)}^{100}{}^{100}C_r(-1)^r(2026)$, satisfies

A) $S<1$ B) $1<S<100$ C) $S>1000$ D) $S>2026$

60. The number of real solutions of the equation $x^3-6x^2+11x-6=0$ is

A) 1 B) 2 C) 3 D) 0

61. Let $p=\tan\left(\frac{5\pi}{9}\cos\left(2\sin^{-1}\frac{1}{\sqrt{5}}\right)\right)$,

$q=\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$

For More Join: @IITJEE_Advanced

then the quadratic equation whose roots are p, sec q is (Here inverse trigonometric functions take principal values)

A) $2\sqrt{3}x^2-6x+\sqrt{3}=0$ B) $\sqrt{3}x^2-4x+2\sqrt{3}=0$

C) $\sqrt{3}x^2-x-2\sqrt{3}=0$ D) $x^2-4\sqrt{3}x+4=0$

62. Find the value of $\lim_{x\to 0}\frac{e^x-x-1}{x^2}$

A) $\frac{1}{2}$ B) 1 C) 2 D) 0

63. If square matrix A is of order 3 then find the value of $|\text{adj}(\text{adj}A)|$, if $|A|=2$

A) 2 B) 4 C) 8 D) 16

64. $f(x)=x^3-x^2f'(1)+xf''(2)-f'''(3), x\in R$ then which of the following is incorrect?

A) $f'(1)=3$ B) $f'(3)=10$ C) $f(0)=-6$ D) $f''(2)=6$

65. **Statement-I:** $f:R\to R$ be a function such that $|f(x)|\leq x^2, \forall x\in R$ then $f(x)$ is differentiable at $x=0$.

Statement-II: $f:R\to R$ be a function such that $|f(x)|\leq x^p, \forall x\in R$ then $f(x)$ is differentiable at $x=0$ for any whole number ' p '.

A) Both statement-I and statement-II are true

B) Both statement-I and statement-II are false

C) Statement-I is true and statement-II is false

D) Statement-I is false and statement-II is true

66. Let $y=f(x)$ satisfy the differential equation $\frac{dy}{dx}=2xe^{-y}, \forall x\in R$. If $y'(1)=1$ then the number of solutions of the equation $f(x)=f'(x)$ in $(0,\infty)$ is

A) 1 B) 2 C) 3 D) zero

For More Join: @IITJEE_Advanced

67. Match the following

	Column-I		ColumnII
(a)	The number of points of discontinuity of $f(x)=(\text{sgn}(x))(\text{sgn}(-x))$ in R is	(i)	0
(b)	The number of points of continuity of $f(x)=\begin{cases} e^{\lvert x\rvert}-1, x \text{ is rational} \\ \frac{1}{1+x^2}, x \text{ is irrational} \end{cases}$ in R is	(ii)	1
(c)	The number of points of discontinuity of $g(x)=\lim_{n\to\infty}\frac{(f(x))^n-1}{(f(x))^n+1}$ in $(0,\infty)$ where $f(x)=(x-1)(x-2)(x-3)+1$ is	(iii)	2
(d)	The number of points of discontinuity of $f(x)f(x)=\max.([x],\lvert x\rvert)$ in R is, where [.] represents the GIF	(iv)	3

Which of the following match is correct?

A) a → iii, b → iv, c → i, d → ii B) a → ii, b → iii, c → iv, d → i

C) a → iii, b → i, c → iv, d → ii D) a → iv, b → i, c → ii, d → iii

68. Let (α,β,γ) be the mirror image of $(1,-2,3)$ in the line $\frac{x+1}{2}=\frac{y-3}{4}=\frac{z+2}{3}$ then $2\alpha+4\beta+3\gamma+2=$

A) 5 B) 11 C) 9 D) 10

69. How many out of the following statements are correct.

STATEMENT I: a,b are non-zero real numbers then $\sqrt{ab}=\sqrt{a}\sqrt{b}$ if at least one of them is positive.

STATEMENT II: a,b are non-zero real numbers then $\sqrt{ab}=\sqrt{a}\sqrt{b}$ if at least one of them is negative

STATEMENT III: a,b are non-zero real numbers then $\sqrt{\frac{a}{b}}=\frac{\sqrt{a}}{\sqrt{b}}$ if $ab>0$ or $(a<0,\ b<0)$

A) 0 B) 1 C) 2 D) 3

70. The number of distinct reflexive relations defined on set A , where set $A=\{1,2,3\}$, is

A) 2^6 B) 2^2 C) 2^{12} D) 2

For More Join: @IITJEE_Advanced

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Find the maximum distance of the point $(-1,-1)$ from the points on circle

$(x-5)^2+(y-7)^2=9$

72. $\sin\beta=\frac{4\sin^2\alpha}{1+4\sin^2\alpha},\sin\gamma=\frac{4\sin^2\beta}{1+4\sin^2\beta},\sin\alpha=\frac{4\sin^2\gamma}{1+4\sin^2\gamma}$ Where $\alpha,\beta,\gamma\in R$, then the maximum value of $\frac{(\sin\alpha+\sin\beta+\sin\gamma)^2}{(\cos\alpha+\cos\beta+\cos\gamma)^2}$ is ?

73. The number of solutions of the equation $\sin^2x+\left(2+2x-x^2\right)\sin x-3(x-1)^2=0$ in $\left[0,\frac{\pi}{2}\right]$ is α, and in $[-2\pi,2\pi]$ is β, then $\alpha+\beta=$ ____

74. If $a_1,a_2,a_3\ldots$ are in Geometric progression, $a_2+a_4+a_6+\ldots..+a_{100}=4$ and $a_1+a_3+a_5+\ldots..+a_{99}=2$ then find the value of constant ratio of the given Geometric progression

75. An online exam is attempted by 40 candidates out of which 15 are boys. The average marks obtained by boys is 10 with variance 2 . The variance of marks obtained by 25 girls is also 2 and the average marks of all 40 candidates is 12.5. If μ is average marks of the 25 girls and σ^2 is variance of marks of all 40 candidates then $20\sigma^2-8\mu=$ ____

For More Join: @IITJEE_Advanced